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Toxicity of seed borne storage fungi of Cowpea (*Vigna sinensis* L.) I- Effect on germination of seed and growth of the radical.

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Abstract : The germination of the seeds of Cowpea due to storage with seed borne *Aspergillus niger, Aspergillus flavus* and *Fusarium moniliforme*, their acetone extract and the metabolite of the noted fungi was recorded considerably suppressed. *A. flavus* proved the most deleterious followed in succession by *A. niger* and *F. moniliforme*. As the relative humidity (RH) of the storage was raised, the suppression of the seed germination was proportionately more due to all the noted fungi. Similar effect was observed on the rate of elongation of the radical due to the metabolite of the noted fungi.

Key Words: Cowpea seeds, storage fungi, germination, growth of radicle, high frequency, respiratory metabolism, depletion, photosynthetic pigments.

INTRODUCTION

The seed borne storage fungi have been reported to deteriorate the seed (Christensen 1957[1], 1972[2]; Harman 1972[3], Bilgrami et al 1979[4], Diwakar and Prasad 1997[5], Prasad 2004[6]). This fate of the seeds have been evaluated opting several parameters such as deviation in colour and texture of the seed coat, suppression of germination, excited activity of respiratory metabolism, depletion of the reserve food, alteration in the enzymatic activity, slow growth of the seedling and less photosynthetic pigments, sugars and amino acids in them, augmented activity of decarboxylase and non-oxidative de aminase of particular amino acid and sluggish activity of nitrate reductase and urease and the mearge amount of the dry weight of the seedlings (Prasad et al 1988[7], Sao *et al* 1989[8], Yadav and Prasad 1997[9], Prasad *et al* 2011[10]).

The present paper deals with the toxicity of seed borne storage fungi of Cowpea with respect to seed germination and the growth of the radical.

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MATERIALS AND METHODS

The storage fungi of Cowpea *(Vigna sinensis* L)seeds collected from farmers/ seed dealers of different Blocks of Ranchi district , were isolated , and based on their high frequency *Aspergillus niger* Van Teighem*, A. flavus* Link ex Fries,and *Fusarium moniliforme* Sheldon were selected to see their effect on the seed germination and the rate of elongation of the radical of their seedlings.

INFESTATION OF THE SEEDS –

The seed fungi were cultured in Czepek Dox Agar slants for seven days at 28± 1^0 C and infested to the seed lot possessing 98% germinability and containing 7.82% moisture worked out earlier. 20 gm of the seed lot was surface sterilized with 0.1% mercuric chloride solution for1 min. This lot was infested with 1ml of the spore suspension of the noted fungi (1 X 10t spore/ml) prepared in 5% Tween 20 and the number was counted with heamocytometer. The infested and uninfested control seed lots were stored over dilute solution of glycerol with distilled water to maintain 60, 70 and 80% RH in sealed desiccators at 30±^{0} C (Brown and Brown 1958[11]) for a period of 30 days.

GERIMATION OF THE SEEDS

The seed lots stored as above were germinated in

autoclaved moist blotters made of three circles of white blotting papers set in the Petri dishes of 8 cm diameter. 20 seeds /blotter in five replicates were set nearly at equal distance and incubated for five days. Triplicate of the five blotters each containing 20 seeds, were used to observe the effect of the storage fungi and the RH of storage. The emergence of radical was considered as germination of the seed. Percentage germination was recorded and presented in table -1.

In another method, similar to the seed lots stored as above, was extracted with 50ml of acetone after surface sterilization of 100 seeds in one replicate, and was filtered on Buchner funnel with two layers of Whatman No. 1 filter paper. The filtrate after centrifugation at 10,000 rpm for 5 minute was evaporated at 50^0C under unreduced pressure. The residue was dissolved in 20ml of distilled water. The seeds in triplicate possessing 98% germinability, were soaked in the extract at 30^0C for 12 hr. These were set for germination as noted earlier. The observation and results of germination was recorded and presented in Table 1.

In still another experiment the noted storage fungi were cultured in 100 ml of Richard solution separately for 10 days at 28 $\pm 1^0$C taking 0.25ml of spore suspension as inoculum. The metabolite was obtained after filtration as noted above for acetone extract maintaining the control of the uninoculated medium. 100 seeds in triplicate were soaked each in 50ml of the metabolites of the fungi and were germinated as noted earlier and percent germination was recorded.

ESTIMATION OF THE RATE OF ELONGATION OF THE RADICLE

100 seeds were soaked in 50ml of the metabolite noted above at 30^0C for 12 hours and germinated as noted earlier. After one day of the emergence of the radicle, the length of 20 of the seedlings taken randomly, were measured in mm of scale daily for five days and the mean length of one radicle was calculated after dividing the total length of twenty radicles by twenty and the results has been recorded and presented in Table 2.

RESULTS AND DISCUSSION –

It appears that the germination of Cowpea seed was suppressed due to all the storage fungi and maximum due to *A. flavus* followed in succession by *A. niger* and *F.moniliforme* (Table 1). Also the germination of infested seeds was suppressed progressively due to rise in RH of the storage. Similar Trend of the seed germination was recorded due to the acetone extract of the fungus stored seeds (Table 1). The control seeds in both the cases had much higher percent of germination than the treated lots. The germination of the seeds was suppressed in the maximum due to the metabolite of *A. flavus* (41% germination), followed in succession by that of *A. niger* (55% germination) and *F. moniliforme* (63% germination) against 98% germination of the control. The rate of increase in the length of radical due to soaking of seeds in the metabolites was clearly sluggish as compare to the control. The trend of detriment of the seed borne fungi was the same as noted for direct storage of the seeds with them, acetone extract of the fungus stored seeds and the metabolite of fungi.

The noted results indicate the deleterious effect of the seed borne storage fungi. Authors have reported the extracellular secretion of toxins in the seed and the culturing medium by these fungi (Harman 1972[3], Harman and Nash 1972[12], Harman and Granett 1972[13])The slow growth of the radical is decisive indication of suppression of cell division and shattering the allied biochemical processes. It is highly conjectured that the metabolite contains the toxic principle of the fungi in the Richard solution that suppresses the seed germination and the rate of elongation of the radical. Singh et al (2009[14]) have recorded distinctly smaller radicle in Bengal gram besides low mitotic index and deviation in the size of the nucleus in the similar condition of storage of the said seed by the same three storage fungi. The presence of toxic principle in the fungus stored seeds is again substantiated by the effect of their acetone extract.

The noted fungi of the seed may also disturb its biochemistry causing damage of the plasma membrane of the seed and consequent leakage of many important cations disturbing the biochemistry at germination and growth of radicle (McKeen and Macdonald 1976[15], Prasad 1980[16],Kishor 1989[17], Sinha 1996[18].). These types of fungi have also been reported to oxidize considerable amount of indole acetic acid by secreting oxidase and moderate its activity causing suppressed growth of the radicle (Kumar 1988[19]).

Table 1—Germination %* of Cowpea (*Vigna sinensis* L.) seeds after their storage with seed borne fungi at varying RH for 30 days and soaking them in Acetone extract.

Particulars	RH (%)				
	Seed borne fungi	60	70	80	Mean
	A.niger	96	88	52	78.6
	A.flavus	95	83	47	75.0
	F.moniliforme	96	90	56	80.6
	CONTROL	98	96	92	95.3
	Mean	96.25	89.25	61.75	
Acetone extract	*A.niger*	97	88	60	81.6
	A.flavus	96	85	53	78.0
	F.moniliforme	96	90	56	80.6
	CONTROL	98	95	90	94.3
	Mean	96.75	89.50	64.74	

*Figures were rounded to their whole number

Table –II : Rate of elongation of the radicle of Cowpea(*Vigna sinensis L.*) raised from the seeds soaked in the metabolite of seed borne fungi in Richard solution (expressed as mm of scale)

Seed borne fungi	Length of the radicle on the day of observation				
	2nd Day	3rd Day	4th Day	5th Day	6Th Day
A.niger	2.1	3.6	5.8	9.2	14.3
A.flavus	2.0	3.2	4.5	6.8	9.4
F.moniliforme	2.4	4.3	6.9	11.4	16.7
CONTROL	3.5	7.1	11.3	17.8	26.4

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